

71253 - Instructions for the final project

Overview

- The goal of the project is for each of you to get hands-on experience in finding a dataset, exploring it to identify interesting questions, analyzing it to answer these questions, and communicating your findings to others in a clear and compelling way.
- Projects are done in teams of 2 students.
- You will need to submit two progress reports, a detailed final report and present your project, as detailed below.

Requirements & Timeline

Due date	Task	Notes
14 April	Find a project partner	Students without a partner will be assigned one randomly
5 May	Submit progress report I	Optional project discussion meeting with course staff
9 June	Submit second progress II report	Optional project discussion meeting with course staff
TBD	Present project	10 min presentation
1 Aug	Submit final report	Late submission will incur a penalty of 2 points per day up to 7 days. Subsequently, submission will not be accepted.

Instructions for progress report I

Each team needs to find a dataset to analyze for their final project, and think about what main questions they would like to address using the data.

What to submit: A short description (1-2 paragraphs) of the dataset you intend to analyze and the main questions you intend to address.

Focus on “big” questions, rather than very specific ones.

That is, ask questions like “What AirBnB should I rent for my next vacation? ” rather than ones like “what is the correlation between an AirBnB’s rating and cost?”).

List the main variables, the number of records, what each record represents. Do not just list the column names (these often mean little to us); explain what information they provide.

Discuss what “issues” that need addressing there are in the data (missing values, inconsistencies, uncertainties, etc’) and what kinds of preprocessing (“cleaning”) of the data may be needed.

Include a link to the source of the data.

The submission should not exceed 1 page.

The submission can be in Hebrew or English.

Instructions for progress report II

Each team needs to finalize the questions they would like to address using the data and think about the proper analysis to answer them.

What to submit: A concise description of the specific questions you intend to address in your data, and of the methods you intend to use to address them.

You can include figures from your exploratory analysis as needed. The submission should not exceed 2 pages, including figures (which should take up most of the space, if you're using all 2 pages).

Instructions for final report

Each team should submit a single report summarizing your final project work. The report should introduce the data you’ve analyzed (including links to the data sources), the exploratory analysis, the specific question/s you’ve addressed, and your overall conclusions. Ideas for follow-up, more in-depth studies (potentially using additional data) are encouraged. The exploratory analysis, before focusing on a specific question, should be a significant part of the project and of the report.

I recommend splitting the analysis section into subsections that each address a specific small question (e.g. are there unusual values in the dataset? Is height dependent on weight?). In each

such subsection, state the exact question, analysis used to address the question, and your conclusions and include the figure/s that best demonstrate your conclusion.

What to submit: A written report in word or pdf format (see specific formatting instructions below). You're welcome to also submit your code, but that is not required and will not be graded.

Formatting instructions:

- Your report should not exceed 10 pages (including figures).
- Minimal font-size: 11
- Minimal line spacing: 1.15
- Minimal margins: 1 inch
- Use section headings to indicate the different sections of your reports (e.g. Introduction, Exploratory Analysis, Identifying factors that predict diabetes, Conclusion).
- Make sure your figures and all the text included in them are large enough to be legible.
- Number your figures. Use these numbers when referring to specific figures in the text. For example: "People find dogs to be 3 times more cute than cats (Figure 3)".
- Give each figure a title stating the main message of the figure.
- Make sure variable names on graphs are informative. You may need to rename columns in your dataframe. (e.g. replace a variable named "m_inc" with "mean_income").
- Do ***NOT*** include code snippets in your report.

FAQs

What is a good dataset to work on for my project?

The most important thing is that you work on a dataset that interests you. That is, a dataset related to a topic where there are questions you'd like to know the answer to.

Technically, the dataset should contain multiple variables (~>10) of different types and multiple records (~>100). You should also have a pretty good understanding of what each variable and record mean.

Note that we will not be learning about time-series analysis, which requires dedicated tools that are outside the scope of this course. Therefore, it is fine to work with data containing a few time points (e.g. comparing weather data from 1980, 2000, and 2020), but not to work with dense time series (e.g. daily weather data).

Additionally, it is discouraged to use a dataset from your own research. These are typically complex datasets whose analysis is outside the scope of this course.

Where can I find a dataset?

The best way is to identify an area/question you are interested in and then search the web for relevant datasets. There are also many sources listing various available datasets. Here are a few of them:

<https://data.gov.il/dataset/>

<https://www.who.int/data/collections>

<https://www.pewresearch.org/datasets/>

<https://data.mendeley.com/>

<https://github.com/awesomedata/awesome-public-datasets>

<https://vincentarelbundock.github.io/Rdatasets/datasets.html>

<https://data.fivethirtyeight.com/>

<https://github.com/owid/owid-datasets/tree/master/datasets> (our world in data; mostly time series)

Note: ** Use of data from [Kaggle.com](https://www.kaggle.com) is not recommended **

While this seems like a great source of datasets, many of the common datasets on this website are not well suited for our course; they are tailored toward very specific questions (mainly prediction ones) and not very amenable to free exploration, which is the intention of this project. In addition, there are many “fake”/synthetic datasets on the website, including highly rated ones. Finally, several of the datasets from this website have been analyzed to death, including by previous students in this course. If you really want to work with a kaggle dataset, talk to us first.

Am I allowed to switch datasets?

Yes. You can decide to change to a different dataset at any point up to your project presentation. Though the later you switch, the less time you have to properly analyze the new dataset. There's no need to ask for permission, though you're welcome to consult with the course staff before deciding to switch.

Can I do the project alone or in a team of more than 2 people?

No. Working in pairs is required.

Does my presentation or final report need to be in English?

No. Your presentations and reports can be in either Hebrew or English. This choice will have no effect on your grade.

Can I get an extension for submitting my final report?

Late submissions without penalty will only be granted under circumstances recognized by university guidelines (e.g. prolonged illness or military reserve duty). Otherwise, late submission will incur a penalty of 2 points per day up to 7 days. Subsequently, submission will not be accepted.